Depth-Supervised Neural Radiance Fields for Accurate Novel View Synthesis

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Abstract—We present Depth-NeRF as a promising solution to address the limitations of Neural Radiance Fields (NeRF). NeRF effectively encodes scenes into neural representations, allowing for the photo-realistic rendering of novel views. However, it falls short in capturing surface geometry, restricting its rendering capabilities to surface colors alone. For overcoming these limitations, we have undertaken substantial modifications to the NeRF network architecture and strategically incorporated depth information to guide its optimization. This method ensures that the ray termination distribution conforms to the scene's geometric properties, thereby producing renderings with improved precision and accuracy. Compared to NeRF, our method achieves a substantial improvement in rendering quality, while simultaneously reducing the RMSE between the rendered and actual scene depth. These improvements unequivocally demonstrate our approach's superior ability to capture the scene's geometric information.

Keywords—NeRF, volume rendering, view synthesis, image-based rendering, depth priors

I. INTRODUCTION

Conventional 3D reconstruction techniques have shown advancement in indoor and outdoor settings, yet they face difficulties in capturing scene details, textures, and changes in illumination. NeRF emerged as a solution, offering superior rendering quality and the ability to synthesize novel views. However, NeRF is not without its limitations, including artifacts and overfitting resulting from a lack of diverse viewpoints. Additionally, the training process is time-consuming, and depth information plays a vital role. To overcome these challenges, our approach utilizes an RGB-D neural radiance field, incorporating both color and depth data for a more comprehensive representation. Our dynamic sampling method efficiently utilizes dense depth information, significantly speeding up the training process while preserving rendering quality.

NeRF, a renowned method [1], has achieved remarkable results with its straightforward approach, utilizing 3D points and view direction as inputs to an MLP for density and color value generation. NeRF, despite its strengths, faces challenges like computational expense and the need for scene-specific models, along with its limitation to static scenes (see [2-4]). DS-NeRF [5] builds upon NeRF by introducing a direct supervision mechanism for the density function. This mechanism uses a loss function to encourage agreement between the ray termination depth distribution and a set of 3D keypoints.

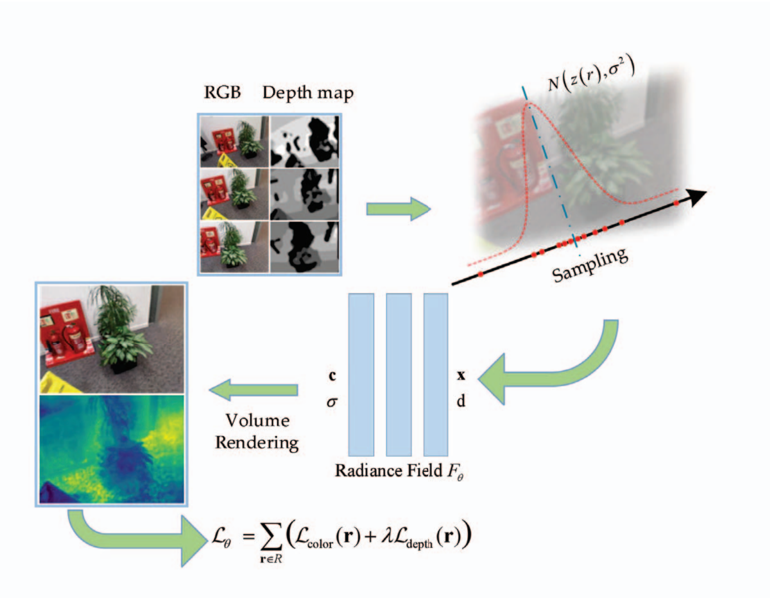

Fig. 1. Overview of our radiance field optimization pipeline.

Reference [6] leverages sparse depth data from Structure-from-Motion (SfM) and converts these points into dense depth information to guide NeRF optimization. The integration of these dense depth priors into NeRF involves a depth constraint and a sampling method. Another method, referred to as "baking" NeRF [7], facilitates real-time rendering on standard hardware. This approach utilizes sparse voxel grids combined with learned feature vectors, this approach facilitates efficient rendering performance while obviating the requirement for access to the primary training dataset. It effectively reformulates the NeRF architecture to enhance performance and speed. To enhance rendering speed, EfficientNeRF [8] introduces both improved sampling strategies for rough and fine stages and a new data structure for caching the entire scene during testing. This caching mechanism significantly accelerates rendering. Enhanced robustness during inference is achieved by randomizing camera positions and directions during training. These improvements lead to more efficient and robust rendering.

Despite considerable progress in view synthesis and the training of NeRF, these techniques still encounter several challenges. DS-NeRF [5], for instance, relies on having a 3D keypoint for each ray, which may be unavailable or difficult to obtain in certain scenarios. Using sparse depth data from Structure-from-Motion (SfM) in some methods can lead to incomplete or inaccurate depth information if the reconstruction process is flawed. DONeRF [9] incorporates an extra depth oracle network during its training process,

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increasing computational time and adding complexity to the training process.

A limitation of EfficientNeRF [8] is its requirement to cache the entire scene at test time, posing a challenge for large-scale scenes or systems with limited memory capacity. Moreover, the method's sensitivity to initial view direction in its sampling approach can affect the consistency and quality of rendered images, potentially leading to biased results.. Surface rendering poses challenges in accurately identifying the precise surface, affecting color consistency across different views and impeding training convergence, Resulting in blurred images, this approach contrasts with volume rendering, which samples numerous points along rays for high-quality results. However, NeRF's per-point evaluation along these rays remains inefficient., resulting in a considerable 22seconds processing time to generate a 640× 480 image on an NVIDIA RTX 3060.

Our method dramatically accelerates NeRF training by using depth information to guide point sampling. By focusing samples around the object surface, we significantly reduce the computational burden. This depth information also serves as a supervisory signal, ensuring accurate light ray termination.

II. METHODOLOGY

A. Volume Rendering with Radiance Fields

Here are a few rewritten versions, each with a slightly different emphasis:

1. More concise explanation:

NeRF uses a neural network to reconstruct 3D scenes from 2D images. This network, called a neural radiance field, maps 3D locations and view directions to density and color values, effectively representing the scene's 3D structure and appearance. By using a collection of posed images, NeRF can reconstruct detailed 3D scenes and enable high-quality rendering from novel viewpoints. More specifically, for a specific point $\mathbf{x} \in R^3$ and a viewing direction $\mathbf{d}$, neural radiance field offers a function that compute the density σ and color $\mathbf{c}$: $(\sigma, \mathbf{c}) = f(\mathbf{x}, \mathbf{d})$.

For image generation from a defined pose $\mathbf{p}$, NeRF emits multiple rays originating at the center of projection $\mathbf{o}$ and proceeding along direction $\mathbf{d}$. Because these rays maintain a constant direction, their propagation distance can be represented as a function of time.: $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$. By sampling points along each ray and passing them through the NeRF network, we obtain density and radiance information at various points along the ray. Integrating these values allows us to estimate the object's color as seen from the camera.

$$\widehat{\mathbf{C}}(\mathbf{r}) = \sum_{k=1}^{K} w_k \mathbf{c}_k \,, \tag{1}$$

$$w_k = T_k \left(1 - \exp\left(-\sigma_k \delta_k\right)\right), \tag{2}$$

$$T_k = \exp\left(-\sum_{k'=1}^{k} \sigma_{k'} \delta_{k'}\right) \tag{3}$$

where $\delta_k = t_{k+1} - t_k$. T_k is transmittance, which is determined by accumulating the density between the start of the ray and point k . w_{k} describes the contribution of each sampled point along the ray to the radiance. NeRF assumes the scene exists within a range $(1, k)$, and to ensure the sum of w_{k} is equal to 1, a non-transparent boundary is placed at k to conceptually constrain the scene. The loss function for NeRF is not defined within this description:

$$\mathcal{L}_{\text{Color}} = \sum_{\mathbf{r} \in \mathcal{R}(\mathbf{p})} \left\| \hat{\mathbf{C}}(\mathbf{r}) - \mathbf{C}(\mathbf{r}) \right\|_2^2 \tag{4}$$

where $\mathcal{R}(\mathbf{p})$ denotes a collection of rays emitted from a stationary pose $\mathbf{p}$.

B. Optimization

Using Equation (1), the surface color $\widehat{\mathbf{C}}(r)$ is computed for each pixel to enhance the optimization of the neural radiance field. The estimated depth $\hat{z}(\mathbf{r})$ and standard deviation $\hat{s}(\mathbf{r})$, produced by NeRF, are used alongside the predicted ray color to inform the radiance field based on dept

$$\hat{z}(\mathbf{r}) = \sum_{k=1}^{K} w_k t_k, \quad \hat{s}(\mathbf{r})^2 = \sum_{k=1}^{K} w_k \left(t_k - \hat{z}(\mathbf{r})\right)^2 \tag{5}$$

To achieve the optimal network parameters θ , we minimize the loss function $\mathcal{L}_\theta$. which is composed of two components: the depth loss function $\mathcal{L}_{\text{depth}}$ and color loss function $\mathcal{L}_{\text{color}}$. Here, λ serves as a hyperparameter that balances the contributions of $\mathcal{L}_{\text{depth}}$ and $\mathcal{L}_{\text{color}}$ in the overall loss.

$$\mathcal{L}_\theta = \sum_{\mathbf{r} \in R} \left(\mathcal{L}_{\text{color}}(\mathbf{r}) + \lambda \mathcal{L}_{\text{depth}}(\mathbf{r})\right) \tag{6}$$

$$\mathcal{L}_{\text{Color}} = \sum_{\mathbf{r} \in \mathcal{R}(\mathbf{p})} \left\| \hat{\mathbf{C}}(\mathbf{r}) - \mathbf{C}(\mathbf{r}) \right\|_2^2 \tag{7}$$

$$\mathcal{L}_{Depth} = \sum_{\mathbf{r} \in \mathcal{R}(\mathbf{p})} \left\| \frac{\left(\hat{z}(\mathbf{r}) - z(\mathbf{r})\right)^2}{\hat{s}(\mathbf{r})^2} \right\|_2^2 \tag{8}$$

By utilizing this method, NeRF is encouraged to conclude rays near the most reliable surface observation informed by the depth prior. At the same time, it maintains a level of adaptability in distributing density to effectively reduce color loss.

	3DMatch rgbd-scenes				ScanNet				3DMatch 7-scenes			
	PSNR ↑	SSIM ↑	LPIPS ↓	Depth-RMSE ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Depth-RMSE ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Depth-RMSE ↓
NeRF	31.55	0.854	0.080	1.113	30.02	0.535	0.208	1.347	30.21	0.810	0.075	1.124
DS-NeRF	29.44	0.613	0.046	0.485	30.03	0.527	0.190	0.092	30.11	0.691	0.122	0.198
NerfingMVS	29.39	0.537	0.274	1.098	29.23	0.546	0.266	0.989	29.62	0.759	0.094	1.163
Ours	**31.97**	**0.863**	0.087	**0.482**	**30.07**	0.532	**0.189**	0.118	**31.36**	0.816	**0.072**	**0.158**

TABLE I. QUANTITATIVE COMPARISON OF THREE REPRESENTATIVE SCENES FROM SCANNET AND 3DMATCH

III. EXPERIMENTS AND ANALYSIS

We evaluated the performance of our algorithm by comparing it with recently proposed approaches. This comparison provided valuable insights into the strengths and weaknesses of our method relative to the latest advancements in the field. Through this evaluation, we validated the effectiveness of our approach and demonstrated its competitiveness in the current research landscape.

A. Experimental Setup

We process rays in batches of 1024 and use the Adam optimizer with a learning rate of 6×10^{-4}. The radiance fields are optimized over 50,000 iterations. We simulate partial depth information loss by randomly perturbing the dataset's depth data.

For a quantitative assessment of view synthesis techniques, we employ several metrics. Newly rendered RGB views are evaluated using peak signal-to-noise ratio (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS). Furthermore, the root-mean-square error (RMSE) is calculated between generated depth maps and the sensor's captured depth data.

B. Basic Comparison

We benchmark our method against NeRF, DS-NeRF, and NerfingMVS.

The data in Table I illustrate that our technique outperforms others across all metrics considered.

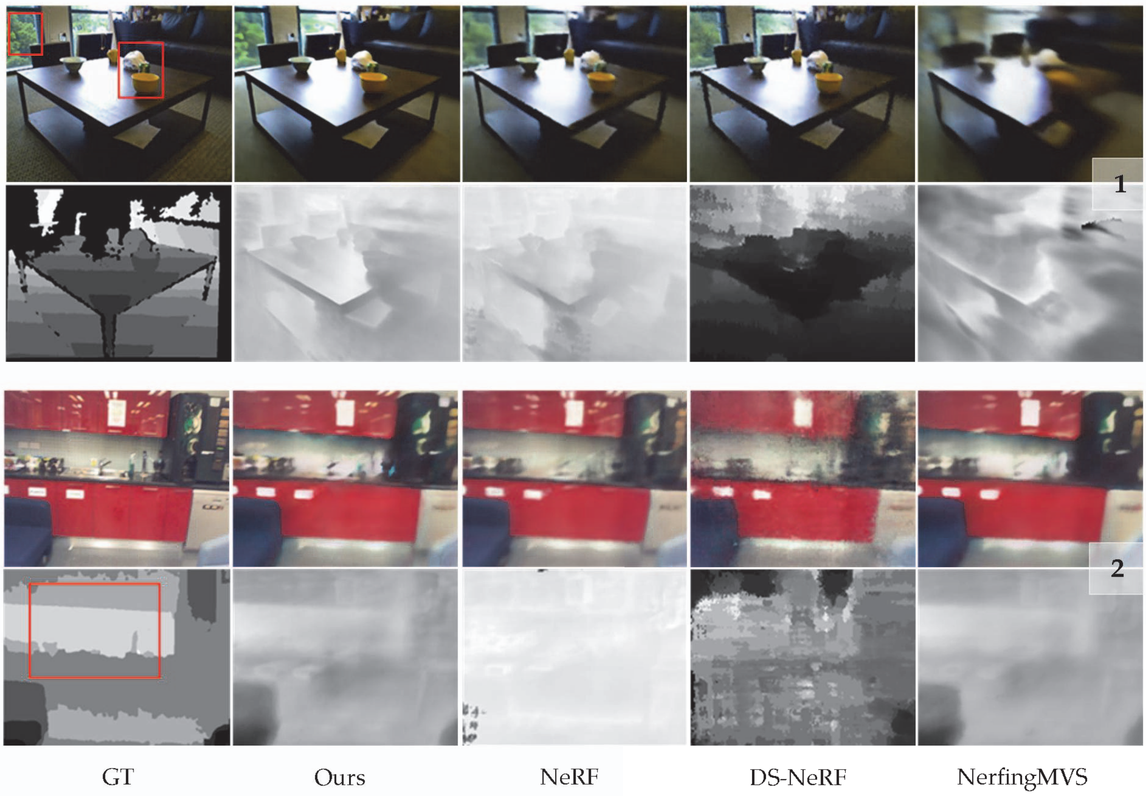

Fig. 2. Comparison between rendered images and depths of the three test scenes and real images and depths

Traditional NeRF approaches often exhibit artifacts when trained on limited input views. Our method mitigates this issue by incorporating dense depth priors and uncertainty, significantly reducing artifacts compared to baselines. Moreover, it improves depth accuracy and color fidelity. Example 1, visualized in Fig. 2, demonstrates the improvements achieved by our method. In contrast to the blurred results of NeRF, our method renders images with enhanced edge definition for the table and other objects. Without depth information, NeRF struggles to establish reliable image correspondences, especially when relying exclusively on color information. This issue is magnified when the number of input images is restricted, contributing to a higher incidence of artifacts in the synthesized views. To address this issue, DS-NeRF [5] tackles the challenge by using the depth of keypoints produced by COLMAP [13] as an additional source of supervision. This additional supervision helps guide the geometry learned by NeRF, resulting in improved performance and reduced artifacts in the final synthesized results.

Although using sparse 3D points poses difficulties, the integration of dense depth priors and uncertainty estimation in our method produces exceptional results, promising broad applicability. The reduction in artifacts, improved depth accuracy, and enhanced color fidelity demonstrate its superior performance. Dual supervision from RGB-derived and 3D point depth information enables reliable correspondences and significantly mitigates artifacts, making our approach a compelling alternative to conventional NeRF methods.

Our method exhibits robustness against outliers present in the sparse depth input. Notably, when erroneous SfM points are present in the cabinets area (as demonstrated in example 2, Fig. 2), alternative approaches suffer more pronounced deficiencies in terms of geometric and color fidelity compared to our proposed method. In contrast, the NerfingMVS approach initially employs COLMAP [13] to obtain depth information and then proceeds to train a monocular depth network specifically tailored to the given scene. The monocular depth network's predicted depth map subsequently serves as a crucial input for guiding and shaping the learning trajectory of the NeRF. Our method's ability to handle outliers in sparse depth data and its reliance on dense depth priors contribute to its superior performance in challenging scenarios, making it a promising solution for accurate 3D scene reconstruction and rendering. Ensuring the accuracy of the predicted depth map is crucial in guaranteeing artifact-free and consistent final rendered images. In our approach, we employ a depth supervision mechanism that selectively focuses on samples near the surface of the scene. This selective approach offers significant benefits during the rendering phase by synthesizing depth information for new viewpoints based on the available depth data. Our method reduces artifacts by aligning ray termination with the scene surface, outperforming baseline methods. The results of this refinement are clearly illustrated in Fig. 2, which demonstrates improved depth accuracy and richer color representation.The improved results stem from our method's outlier management, customized training strategy, and selective integration of depth information during rendering. These advancements produce high-fidelity images, making our approach a promising solution for accurate and visually compelling 3D scenes.

C. Ablation Study

The effectiveness of the individual components was assessed through ablation studies on the ScanNet and 3DMatch datasets. Quantitative results in Table 2 and qualitative examples in Figure 3 demonstrate the superior performance of our full method in terms of both image quality and depth estimation, further confirming the superiority of our approach in producing high-quality rendered images and accurate depth estimations. The comprehensive evaluation on multiple datasets provides robust evidence supporting the efficacy of our method's added components and their positive impact on the overall performance.

The absence of deep supervision introduces shape-radiance ambiguity. Although our approach generates high-quality RGB images from new perspectives, accurately representing the scene's geometry remains challenging. Only a small portion of the generated depth map contains accurate depth information, while the differences in depth across the rest of the areas are insignificant, making it insufficient for capturing the true depth of the scene. This limitation hinders our ability to achieve a complete and precise representation of the scene's geometry despite the visually appealing rendered images.

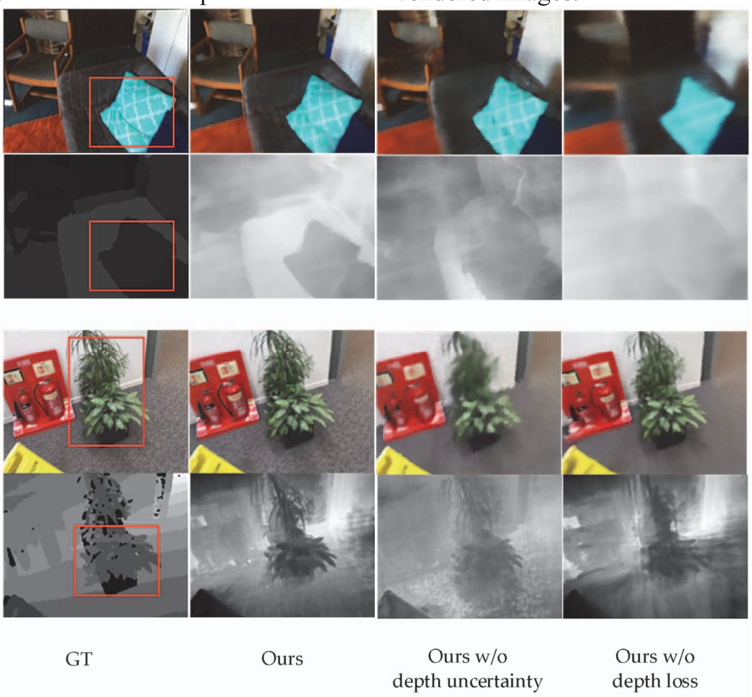

Fig. 3. Rendered RGB and depth images for test views from ScanNet and 3DMatch

TABLE II. QUANTITATIVE ANALYSIS OF THREE REPRESENTATIVE SCENES

	3DMatch 7-scenes-fire				ScanNet scene0002_00			
	PSNR ↑	SSIM ↑	LPIPS ↓	Depth-RMSE ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Depth-RMSE ↓
Depth-NeRF w/o uncertainty	29.77	0.548	0.037	0.172	28.54	0.495	0.277	0.203
Depth-NeRF w/o depth loss	30.30	0.571	0.098	1.114	29.58	0.533	0.321	0.523
Depth-NeRF	**30.42**	**0.592**	0.104	**0.152**	**30.03**	**0.566**	**0.186**	**0.201**

Eliminating uncertainty from the optimization process leads to challenges in resolving inconsistencies in the overlapping regions of the 2D depth priors. This, in turn, gives rise to erroneous edges in both RGB and depth images (as observed in example 1, Fig. 3), duplication artifacts (as evident in example 2, Fig. 3).

The quantitative results on the ScanNet dataset (Table. II) emphasize the significance of considering uncertainty, particularly when dealing with lower quality sparse depth data from Structure-from-Motion (SfM). Incorporating uncertainty becomes crucial in addressing these inconsistencies and ensuring the production of accurate and high-quality reconstructions, especially when the input depth information is of lower fidelity.

IV. CONCLUSION

We have introduced a novel view synthesis method using neural radiance fields (NeRF) that harnesses the power of dense depth priors. Unlike other NeRF techniques that rely on Structure-from-Motion (SfM) for depth information, our method directly extracts depth from RGB-D images. This approach eliminates the need for sufficient translational motion, which can be a disadvantage when using SfM, as it requires careful initialization and may fail in certain scenarios, limiting the wider applicability of NeRF techniques. By incorporating dense depth priors with uncertainty during NeRF optimization. Compared to other methods, ours provides significantly better novel view image quality and more accurate depth estimates. Integrating depth information into NeRF training enhances rendering quality and provides continuous scene depth, making NeRF reconstructions more accessible and applicable. This advancement addresses the limitations of current techniques and paves the way for more effective and practical applications of NeRF in view synthesis and 3D scene reconstruction.

REFERENCES

[1] Mildenhall, B.; Srinivasan, P.P.; Tancik, M.; Barron, J.T.; Ramamoorthi, R.; Ng, R. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis 2020.

[2] Gafni, G.; Thies, J.; Zollhofer, M.; Niesner, M. Dynamic Neural Radiance Fields for Monocular 4D Facial Avatar Reconstruction. In Proceedings of the 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: Nashville, TN, USA, June 2021; pp. 8645–8654.

[3] Tretschk, E.; Tewari, A.; Golyanik, V.; Zollhofer, M.; Lassner, C.; Theobalt, C. Non-Rigid Neural Radiance Fields: Reconstruction and Novel View Synthesis of a Dynamic Scene From Monocular Video. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV); IEEE: Montreal, QC, Canada, October 2021; pp. 12939–12950.

[4] Pumarola, A.; Corona, E.; Pons-Moll, G.; Moreno-Noguer, F. D-NeRF: Neural Radiance Fields for Dynamic Scenes. In Proceedings of the 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: Nashville, TN, USA, June 2021; pp. 10313–10322.

[5] Deng, K.; Liu, A.; Zhu, J.-Y.; Ramanan, D. Depth-Supervised NeRF: Fewer Views and Faster Training for Free. In Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: New Orleans, LA, USA, June 2022; pp. 12872–12881.

[6] Roessle, B.; Barron, J.T.; Mildenhall, B.; Srinivasan, P.P.; Niebner, M. Dense Depth Priors for Neural Radiance Fields from Sparse Input Views. In Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: New Orleans, LA, USA, June 2022; pp. 12882–12891.

[7] Hedman, P.; Srinivasan, P.P.; Mildenhall, B.; Barron, J.T.; Debevec, P. Baking Neural Radiance Fields for Real-Time View Synthesis. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV); IEEE: Montreal, QC, Canada, October 2021; pp. 5855–5864.

[8] Hu, T.; Liu, S.; Chen, Y.; Shen, T.; Jia, J. EfficientNeRF Efficient Neural Radiance Fields.

[9] Neff, T.; Stadlbauer, P.; Parger, M.; Kurz, A.; Mueller, J.H.; Chaitanya, C.R.A.; Kaplanyan, A.; Steinberger, M. DONeRF: Towards Real-Time Rendering of Compact Neural Radiance Fields Using Depth Oracle Networks. *Comput. Graph. Forum* **2021**, *40*, 45–59, doi:10.1111/cgf.14340.

[10] Wang, Z.; Bovik, A.C.; Sheikh, H.R.; Simoncelli, E.P. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Trans. Image Process.* **2004**, *13*, 600–612, doi:10.1109/TIP.2003.819861.

[11] Zhang, R.; Isola, P.; Efros, A.A.; Shechtman, E.; Wang, O. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition; IEEE: Salt Lake City, UT, June 2018; pp. 586–595.

[12] Wei, Y.; Liu, S.; Rao, Y.; Zhao, W.; Lu, J.; Zhou, J. NerfingMVS: Guided Optimization of Neural Radiance Fields for Indoor Multi-View Stereo. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV); IEEE: Montreal, QC, Canada, October 2021; pp. 5590–5599.

[13] Schonberger, J.L.; Frahm, J.-M. Structure-from-Motion Revisited. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: Las Vegas, NV, USA, June 2016; pp. 4104–4113.

[14] Dai, A.; Chang, A.X.; Savva, M.; Halber, M.; Funkhouser, T.; Niessner, M. ScanNet: Richly-Annotated 3D Reconstructions of Indoor Scenes. In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: Honolulu, HI, July 2017; pp. 2432–2443.

[15] Zeng, A.; Song, S.; Niessner, M.; Fisher, M.; Xiao, J.; Funkhouser, T. 3DMatch: Learning Local Geometric Descriptors from RGB-D Reconstructions. In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); IEEE: Honolulu, HI, July 2017; pp. 199–208.